

MTH 166

Lecture-29

**Parametric Equation of a Tangent Line,
Length of Space Curve, Motion of a Particle**

Topic:

Vector Differential Calculus

Learning Outcomes:

1. To find parametric equation of tangent line
2. To find Length of Space Curve
3. Motion of a body or a particle

Parametric Equation of Tangent Line:

Problem 1. Find the parametric equation of tangent line to the curve:

$$x = t, y = 2t^2, z = 3t^3 \text{ at } t = 2.$$

Solution. Let $P_0 = (x, y, z)_{t=2} = (t, 2t^2, 3t^3)_{t=2} = (2, 8, 24)$

$$\text{Let } \vec{v} = \left(\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} \right)_{t=2} = (\hat{i} + 4t\hat{j} + 9t^2\hat{k})_{t=2} = (\hat{i} + 8\hat{j} + 36\hat{k})$$

Parametric equation of tangent line to the given curve is:

$$\overrightarrow{r(\mu)} = \overrightarrow{P_0} + \mu \vec{v} \quad \text{where } \mu \text{ is parameter}$$

$$\Rightarrow \overrightarrow{r(\mu)} = (2\hat{i} + 8\hat{j} + 24\hat{k}) + \mu(\hat{i} + 8\hat{j} + 36\hat{k})$$

$$\Rightarrow \overrightarrow{r(\mu)} = (2 + \mu)\hat{i} + (8 + 8\mu)\hat{j} + (24 + 36\mu)\hat{k} \quad \textbf{Answer.}$$

Problem 2. Find the parametric equation of tangent line to the curve:

$$x = t, y = e^t, z = 1 \text{ at } t = 1.$$

Solution. Let $P_0 = (x, y, z)_{t=1} = (t, e^t, 1)_{t=1} = (1, e, 1)$

$$\text{Let } \vec{v} = \left(\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} \right)_{t=1} = (\hat{i} + e^t \hat{j} + 0 \hat{k})_{t=1} = (\hat{i} + e \hat{j})$$

Parametric equation of tangent line to the given curve is:

$$\overrightarrow{r(\mu)} = \overrightarrow{P_0} + \mu \vec{v} \quad \text{where } \mu \text{ is parameter}$$

$$\Rightarrow \overrightarrow{r(\mu)} = (\hat{i} + e \hat{j} + \hat{k}) + \mu(\hat{i} + e \hat{j})$$

$$\Rightarrow \overrightarrow{r(\mu)} = (1 + \mu)\hat{i} + (e + e\mu)\hat{j} + \hat{k} \quad \textbf{Answer.}$$

Length of Space Curve:

Let the curve C be represented in the parametric form as:

$$\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}, \quad a \leq t \leq b$$

Then, length of curve C is given by:

$$l = \int_{t=a}^{t=b} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Problem 1. Find the length of the following curve:

$$\overrightarrow{r(t)} = (a \cos t)\hat{i} + (a \sin t)\hat{j}, \quad 0 \leq t \leq 2\pi$$

Solution. Comparing with: $\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j}, \quad a \leq t \leq b$

$$x(t) = a \cos t \quad \Rightarrow \frac{dx}{dt} = -a \sin t \quad \text{and} \quad y(t) = a \sin t \quad \Rightarrow \frac{dy}{dt} = a \cos t$$

Length of curve C is given by: $l = \int_{t=a}^{t=b} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

$$\Rightarrow l = \int_{t=0}^{t=2\pi} \sqrt{(-a \sin t)^2 + (a \cos t)^2} dt = a \int_{t=0}^{t=2\pi} dt = a(2\pi - 0)$$

$$\Rightarrow l = 2\pi a \text{ units} \quad \mathbf{Answer.}$$

Problem 2. Find the length of the following curve:

$$\overrightarrow{r(t)} = (\cos t)\hat{i} + (\sin t)\hat{j} + (3t)\hat{k}, \quad -2\pi \leq t \leq 2\pi$$

Solution. Comparing with: $\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}, \quad a \leq t \leq b$

$$x(t) = \cos t \Rightarrow \frac{dx}{dt} = -\sin t, \quad y(t) = \sin t \Rightarrow \frac{dy}{dt} = \cos t, \quad z(t) = 3t \Rightarrow \frac{dz}{dt} = 3$$

$$\text{Length of curve } C \text{ is given by: } l = \int_{t=a}^{t=b} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

$$\Rightarrow l = \int_{t=-2\pi}^{t=2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (3)^2} dt = \int_{t=-2\pi}^{t=2\pi} \sqrt{1 + 9} dt$$

$$\Rightarrow l = \sqrt{10}(2\pi + 2\pi) = 4\pi\sqrt{10} \text{ units} \quad \textbf{Answer.}$$

Motion of a body or a particle

Let position vector of a particle be given by:

$$\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

Then, we can calculate following quantities:

1. Velocity, $\overrightarrow{V(t)} = \frac{d}{dt} (\overrightarrow{r(t)})$

2. Speed, $S = |\overrightarrow{V(t)}|$ (Magnitude of vector $\overrightarrow{V(t)}$)

3. Acceleration, $\overrightarrow{a(t)} = \frac{d}{dt} (\overrightarrow{V(t)})$

Problem 1. The position vector of a moving particle is given by:

$\overrightarrow{r(t)} = (\cos t + \sin t)\hat{i} + (\sin t - \cos t)\hat{j} + t\hat{k}$. Determine the velocity, speed and acceleration of the particle in the direction of the motion.

Solution. $\overrightarrow{r(t)} = (\cos t + \sin t)\hat{i} + (\sin t - \cos t)\hat{j} + t\hat{k}$

Velocity, $\overrightarrow{V(t)} = \frac{d}{dt}(\overrightarrow{r(t)}) = \frac{d}{dt}[(\cos t + \sin t)\hat{i} + (\sin t - \cos t)\hat{j} + t\hat{k}]$

$\Rightarrow \overrightarrow{V(t)} = (-\sin t + \cos t)\hat{i} + (\cos t + \sin t)\hat{j} + 1\hat{k}$

Speed, $S = |\overrightarrow{V(t)}| = \sqrt{(-\sin t + \cos t)^2 + (\cos t + \sin t)^2 + (1)^2}$

$\Rightarrow S = \sqrt{2(\sin^2 t + \cos^2 t) + 1} = \sqrt{3}$

Acceleration, $\overrightarrow{a(t)} = \frac{d}{dt}(\overrightarrow{V(t)}) = \frac{d}{dt}[(-\sin t + \cos t)\hat{i} + (\cos t + \sin t)\hat{j} + 1\hat{k}]$

$\Rightarrow \overrightarrow{a(t)} = (-\cos t - \sin t)\hat{i} + (-\sin t + \cos t)\hat{j} + 0\hat{k}$

Problem 2. The position vector of a moving particle is given by:

$\overrightarrow{r(t)} = t^3\hat{i} + t\hat{j} + t^2\hat{k}$. Determine the velocity, speed and acceleration of the particle in the direction of the motion.

Solution. $\overrightarrow{r(t)} = t^3\hat{i} + t\hat{j} + t^2\hat{k}$

Velocity, $\overrightarrow{V(t)} = \frac{d}{dt}(\overrightarrow{r(t)}) = \frac{d}{dt}[t^3\hat{i} + t\hat{j} + t^2\hat{k}]$

$$\Rightarrow \overrightarrow{V(t)} = 3t^2\hat{i} + \hat{j} + 2t\hat{k}$$

Speed, $S = |\overrightarrow{V(t)}| = \sqrt{(3t^2)^2 + (1)^2 + (2t)^2}$

$$\Rightarrow S = \sqrt{9t^4 + 4t^2 + 1}$$

Acceleration, $\overrightarrow{a(t)} = \frac{d}{dt}(\overrightarrow{V(t)}) = \frac{d}{dt}[3t^2\hat{i} + \hat{j} + 2t\hat{k}]$

$$\Rightarrow \overrightarrow{a(t)} = 6t\hat{i} + 0\hat{j} + 2\hat{k}$$

Rules of Vector Differentiation

If $\vec{u}(t), \vec{v}(t)$ are vector functions and α, β are constants, then following rules of differentiation hold for vector functions in same way as that for scalar function.

$$1. \frac{d}{dt} [\alpha \vec{u} \pm \beta \vec{v}] = \alpha \frac{d}{dt} (\vec{u}) \pm \beta \frac{d}{dt} (\vec{v})$$

$$2. \frac{d}{dt} [\vec{u} \cdot \vec{v}] = \vec{u} \frac{d}{dt} (\vec{v}) + \vec{v} \frac{d}{dt} (\vec{u}) \quad \left(\frac{d}{dt} (\text{Dot product between } \vec{u} \text{ and } \vec{v}) \right)$$

$$3. \frac{d}{dt} [\vec{u} \times \vec{v}] = \vec{u} \times \frac{d}{dt} (\vec{v}) + \vec{v} \times \frac{d}{dt} (\vec{u}) \quad \left(\frac{d}{dt} (\text{Cross product between } \vec{u} \text{ and } \vec{v}) \right)$$

$$4. \frac{d}{dt} \left[\frac{\vec{u}}{\vec{v}} \right] = \frac{\vec{v} \frac{d}{dt} (\vec{u}) - \vec{u} \frac{d}{dt} (\vec{v})}{(\vec{v})^2}, \quad \text{provided } \vec{v} \neq 0$$

Problem 1. If $\overrightarrow{u(t)} = (\sin 2t)\hat{i} - (\cos 2t)\hat{j} + t\hat{k}$, $\overrightarrow{v(t)} = (\cos 2t)\hat{i} - (\sin 2t)\hat{j} + t^2\hat{k}$

Find $[\overrightarrow{u(t)} \cdot \overrightarrow{v(t)}]'$

Solution. Here we will first find dot product between $\overrightarrow{u(t)}$ and $\overrightarrow{v(t)}$

$$\begin{aligned}\text{i.e. } [\overrightarrow{u(t)} \cdot \overrightarrow{v(t)}] &= [(\sin 2t)\hat{i} - (\cos 2t)\hat{j} + t\hat{k}] \cdot [(\cos 2t)\hat{i} - (\sin 2t)\hat{j} + t^2\hat{k}] \\ &= (\sin 2t)(\cos 2t) + (-\cos 2t)(-\sin 2t) + (t)(t^2) \\ &= \sin 4t + t^3\end{aligned}$$

$$\begin{aligned}\text{So, } [\overrightarrow{u(t)} \cdot \overrightarrow{v(t)}]' &= \frac{d}{dt} [\overrightarrow{u(t)} \cdot \overrightarrow{v(t)}] = \frac{d}{dt} [\sin 4t + t^3] \\ &= 4 \cos 4t + 2t^2 \quad \textbf{Answer.}\end{aligned}$$

Problem 2. If $\overrightarrow{u(t)} = (6t^2)\hat{i} - (t)\hat{j} + (3t^2)\hat{k}$, $\overrightarrow{v(t)} = (t)\hat{i} + (t^2)\hat{j} + (2t)\hat{k}$

Find $[\overrightarrow{u(t)} \times \overrightarrow{v(t)}]'$

Solution. Here we will first find cross product between $\overrightarrow{u(t)}$ and $\overrightarrow{v(t)}$

$$\text{i.e. } [\overrightarrow{u(t)} \times \overrightarrow{v(t)}] = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6t^2 & -t & 3t^2 \\ t & t^2 & 2t \end{vmatrix} = -(2t^2 + 3t^4)\hat{i} - 9t^3\hat{j} + (6t^4 + t^2)\hat{k}$$

$$\begin{aligned} \text{So, } [\overrightarrow{u(t)} \times \overrightarrow{v(t)}]' &= \frac{d}{dt} [\overrightarrow{u(t)} \times \overrightarrow{v(t)}] = \frac{d}{dt} [-(2t^2 + 3t^4)\hat{i} - 9t^3\hat{j} + (6t^4 + t^2)\hat{k}] \\ &= -(4t + 12t^3)\hat{i} - 27t^2\hat{j} + (24t^3 + 2t)\hat{k} \textbf{Answer.} \end{aligned}$$

Problem 3. Prove that $\left[\overrightarrow{v(t)} \times \overrightarrow{v'(t)}\right]' = \overrightarrow{v(t)} \times \overrightarrow{v''(t)}$

Solution. Here $\left[\overrightarrow{v(t)} \times \overrightarrow{v'(t)}\right]' = \frac{d}{dt} \left[\overrightarrow{v(t)} \times \overrightarrow{v'(t)}\right]$

$$\begin{aligned} &= \overrightarrow{v(t)} \times \frac{d}{dt} \left(\overrightarrow{v'(t)}\right) + \overrightarrow{v'(t)} \times \frac{d}{dt} \left(\overrightarrow{v(t)}\right) \\ &= \left[\overrightarrow{v(t)} \times \overrightarrow{v''(t)}\right] + \left[\overrightarrow{v'(t)} \times \overrightarrow{v'(t)}\right] \\ &= \left[\overrightarrow{v(t)} \times \overrightarrow{v''(t)}\right] + 0 \quad (\text{Cross product of a vector with itself is always zero}) \\ &= \overrightarrow{v(t)} \times \overrightarrow{v''(t)} \end{aligned}$$

Hence proved.

